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Customizable waveguides and associated systems and methods

Abstract

Customizable waveguides for use in a loudspeaker and associated systems and methods are disclosed herein. In some embodiments, the waveguide includes a frame and a waveguide insert sealably couplable to the frame. The frame can include a first end region that can be coupled to a compression driver in the loudspeaker and a second end region opposite the first end region. The second end region can be coupled to a front baffle of the loudspeaker when the first end region is coupled to the compression driver. Further, the frame can be symmetric about vertical and/or horizontal axes of the frame. The waveguide insert can include a first half-horn couplable to a first longitudinal half of the frame, a second half-horn couplable to a second longitudinal half of the frame opposite the first longitudinal half, and a divider positioned between the first half-horn and the second half-horn.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to speaker technology. More specifically, the present technology relates to customizable waveguides to shape acoustic waves from loudspeakers.

BACKGROUND

(2) Loudspeakers typically use a waveguide (also referred to as a speaker horn) to improve the overall efficiency of the driving element in the speaker and to direct the resulting acoustic wave toward a target. For example, a typical loudspeaker has a compression driver that oscillates to produce sound waves. The compression driver is attached to a waveguide that improves the coupling between the compression driver and the surrounding air, for example by providing a form of impedance matching between the material of the compression driver and the air. To do so, waveguides have a narrow section (referred to as the “throat”) coupled to the compression driver, a middle section (referred to as the “neck”) that gradually expands to shape and expand the acoustic wave, and an end section (referred to as the “mouth”) that projects the acoustic wave out of the speaker (e.g., is the part of the horn that interacts with external ambient air). As a result of the wave-shaping and impedance matching, the waveguide can significantly improve the sound output from the compression driver.

(3) The shape of the neck can also impact the direction of the acoustic wave exiting the waveguide. Modern loudspeakers are manufactured with specific angles in their waveguides to better direct acoustic waves into the spaces that the loudspeakers are used in. For example, a wide-angle loudspeaker can be used to direct the acoustic waves generally into a large room (e.g., a conference room, concert hall, movie theater, and the like), while a narrow-angle loudspeaker can be used to fill in specific spaces in the room that are not well covered by the wide-angle loudspeaker.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic top view of a space employing loudspeakers with various beamwidths tailored to the space.
- (2) FIG. 2 is an partially schematic view of a loudspeaker configured in accordance with embodiments of the present technology.
- (3) FIGS. 3A and 3B are exploded partially schematic views of a customizable waveguide configured in accordance with embodiments of the present technology.
- (4) FIG. 4 is a partial cross-sectional view of a customizable waveguide configured in accordance with embodiments of the present technology.
- (5) FIG. 5A is a partially schematic view of a frame for a customizable waveguide configured in accordance with embodiments of the present technology.
- (6) FIG. 5B is an enlarged rear, partially schematic view of the frame of FIG. 5A.
- (7) FIG. 6 is a partial cross-sectional view of a customizable waveguide configured in accordance with further embodiments of the present technology.
- (8) FIG. 7 is a schematic top view of a space employing loudspeakers with customizable waveguides configured in accordance with embodiments of the present technology.
- (9) The drawings have not necessarily been drawn to scale. Similarly, some components and/or operations can be separated into different blocks or combined into a single block for the purpose of

discussion of some of the implementations of the present technology. Moreover, while the technology is amenable to various modifications and alternative forms, specific implementations have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the technology to the particular implementations described.

DETAILED DESCRIPTION

Overview

(10) Loudspeaker systems can be manufactured with highly specialized waveguides and specific loudspeaker placement to create complex acoustic effects (e.g., surround sound, 3D sound environments, and the like) for spaces such as conference centers, concert halls, movie theaters, churches, and the like. More specifically, the loudspeakers can be created and placed in specific places to ensure complete acoustic coverage of the space with relatively few overlaps and/or dead zones. The overlaps can result in various distortions of the sound that undermine artistic intent and/or a listener's experience. Similarly, the dead zones can create pockets in the space that are not ideal for listeners in the audience. However, the loudspeakers created for these environments can be limited in use since they are specifically created for a particular environment and/or experience. Further, they may require particular placements of the loudspeakers that can be difficult to work around as an environment changes (e.g., as a conference center is altered between different conferences).

(11) Loudspeakers having customizable waveguides and associated systems and methods that address these shortcomings are disclosed herein. A representative loudspeaker can include a compression driver (or other suitable driver, such as a piezo driver, a dome driver, a cone driver, and/or the like) positioned within a loudspeaker housing and directed toward a front baffle of the housing. The waveguide can include a frame operably couplable (or sealably coupled) between the compression driver and the front baffle, as well as a waveguide insert sealably couplable to the frame. The operable coupling can help direct any acoustic waves generated by the compression driver into the frame and toward the front baffle. The operable coupling can also secure the frame in position such that the acoustic waves do not vibrate (or otherwise move) the frame, which could result in distortion of the acoustic waves. A sealable coupling can help ensure that there are no (or very few) interfaces with ambient air (e.g., non-driven air within the speaker housing) through the frame and/or the waveguide insert. The sealable coupling can help allow the waveguide insert to shape an acoustic wave and perform any necessary impedance matching between the compression driver and the front baffle. As discussed in more detail below, the frame can provide a fixed structure while the waveguide insert can be customizable for a specific setting and/or use of the loudspeaker. Furthermore, the waveguide insert can be asymmetric about one or more axes and/or rotatable within the frame. As a result, the waveguide insert can allow a single loudspeaker to be configured to produce multiple different acoustic coverage angles.

(12) The frame can include a first end region (e.g., a throat) that can be coupled to the compression driver in the loudspeaker and a second end region (e.g., a mouth) that can be coupled to the front baffle of the loudspeaker. In some embodiments, the frame can be symmetric about the vertical and/or horizontal axes of the frame. Purely by way of example, the mouth of the frame can have a generally square perimeter, allowing the waveguide insert to be rotated by 90 degrees to achieve varying coverages.

(13) In some embodiments, the waveguide insert includes a first half-horn couplable to a first longitudinal half of the frame, a second half-horn couplable to a second longitudinal half of the frame, and a divider positionable between the first half-horn and the second half-horn. The first half-horn can have a first acoustic beamwidth while the second half-horn has a second acoustic beamwidth. The first and second beamwidths can be different, resulting in asymmetric coverage from the waveguide insert customizable to a specific space and/or intended use for the loudspeaker. For example, in some embodiments, the first half-horn and/or the second half-horn can be switched with various other half-horns with various other beamwidths. For example, the waveguide insert

can be decoupled from the frame, the first half-horn and/or the second half-horn can be swapped with another half-horn, and the waveguide insert can be recoupled to the frame. The alterations can customize the acoustic coverage from the waveguide insert as needed for a space and/or specific use of the loudspeaker. Additionally, or alternatively, the waveguide insert can be decoupled from the frame, rotated about a longitudinal axis, and recoupled to the frame in a new orientation.

(14) For ease of reference, loudspeakers, waveguides, and components thereof are sometimes described herein with reference to top and bottom, upper and lower, upwards and downwards, and/or horizontal plane, x-y plane, vertical, or z-direction relative to the spatial orientation of the embodiments shown in the figures. It is to be understood, however, that the loudspeakers, waveguides, and components thereof can be moved to, and used in, different spatial orientations without changing the structure and/or function of the disclosed embodiments of the present technology.

DESCRIPTION OF THE FIGURES

(15) FIG. 1 is a schematic top view of a space **101** employing a loudspeaker system **110** with various acoustic beamwidths. In various embodiments, the space **101** can be a conference room, a concert hall, a movie theater, and/or any various other rooms (or outdoor spaces) in which loudspeakers are utilized to project sound (speeches, commentary, dialog, music, movies, or other audible output), for example, to an audience having one or more recipients. The space **101** includes a target area **102** in which the audience would likely be positioned. As a result, the loudspeaker system **110** can be positioned and configured to generally target the target area **102**.

(16) In the configuration illustrated in FIG. 1, providing the loudspeaker system **110** for use in the space **101** provides significant challenges, particularly when the system has loudspeakers **112** mounted adjacent to a wall or ceiling that can impede or interfere with propagation of the sound toward the target area **102** in the space **101**. For example, each loudspeaker **112** can include a waveguide **115** that receives sound waves generated by a driving element **114**, such as a compression driver, a piezo driver, a dome driver, a cone driver, and/or the like. The waveguide **115** can be mounted to a front baffle **116** of the loudspeaker **112**. The waveguide **115** (sometimes also referred to herein as a “speaker horn,” a “horn waveguide,” and/or the like) directs the acoustic waves away from the loudspeaker **112** toward the target area **102**. When the loudspeaker **112** has a substantially symmetrical waveguide and is mounted adjacent to a wall, a ceiling, and/or other barrier, the waveguide **115** will produce a beamwidth initially symmetrical about a primary plane. For example, as illustrated by a first sound path **120** in FIG. 1, the waveguide **115** can shape the acoustic wave to exit the loudspeaker **112** at a first angle A off a plane of the front baffle **116** (which is, for example, perpendicular to the primary plane) to both the left and right (in the illustrated orientation) of the loudspeaker **112**. As a result, as illustrated by the first sound path **120**, each of the loudspeakers **112** can provide good acoustic coverage toward the center of the target area **102**. However, the first sound path **120** will reflect off the adjacent wall of the space **101**, thereby resulting in distortions to the sound, as well as areas with double coverage, and areas with only distorted coverage. In this configuration, the loudspeakers **112** do not resolve the challenges present when the loudspeaker is mounted next to the wall (or other interfering barrier), and the result is a suboptimal sound propagation to the target area **102** and also poor sound coverage to the sides of the target area **102**.

(17) Some loudspeakers **112** can be specifically constructed with a fixed asymmetric waveguide **115** to better suit the target area **102**. For example, the customized waveguide **115** can have a beamwidth asymmetrical about the primary plane. For example, as illustrated by a second sound path **130**, the customized waveguide **115** can shape the acoustic wave to exit the loudspeakers **112** at a second angle B toward the center of the target area **102** and a third angle C toward the sides of the target area **102**. The second and third angles B and C can be set based on where the loudspeakers **112** are placed in the space **101** and the dimensions of the target area **102** in the space **101**. As a result, the second sound path **130** provides better coverage over the target area **102** than

the first sound path **120**. Such loudspeakers with the specifically shaped waveguide **115** to fit the particular target area **102** and/or the space **101**, however, have significant limitations if the loudspeakers were to be used in a different space **101** with a different target area **102**.

(18) FIG. 2 is a partially schematic view of a loudspeaker **200** configured in accordance with one or more embodiments of the present technology that can solve the challenges experienced by sound systems for installation in a wide range of spaces with different configurations. In the illustrated embodiment, the loudspeaker **200** includes a housing **202** with a front baffle **204**. The loudspeaker **200** also includes a bass component **210** (e.g., a woofer or subwoofer component) and a horn-speaker component **220** each coupled to the front baffle **204**. The bass component **210** can generate acoustic waves with a relatively low-frequency range (e.g., about 60-500 Hertz (Hz)) while the horn-speaker component **220** can generate acoustic waves with a midrange to high-frequency range (e.g., about 500 Hz-20 kHz).

(19) In the illustrated embodiment, the horn-speaker component **220** includes a compression driver **222** (or other suitable driving element) and a customizable waveguide **230**. The waveguide **230** has a frame **232** coupled between the compression driver **222** and the front baffle **204**. The waveguide **230** also includes a waveguide insert **240** sealably and removably coupled to the frame **232**. The waveguide insert **240** (sometimes also referred to herein as a “customizable waveguide system”) receives, shapes, and directs acoustic waves generated by the compression driver **222**. Meanwhile, the frame **232** provides structural and mounting support for the waveguide insert **240**. The frame **232** can help ensure that the waveguide insert **240** remains rigidly fixed to the frame **232** and within the speaker's housing **202** to avoid creating distortions in the acoustic waves. As discussed in more detail below, the waveguide insert **240** having one shape or configuration can be easily and quickly removed from the frame **232** and replaced with another waveguide insert **240** with a different shape or configuration (and/or components of the waveguide insert **240** can be updated and/or changed to create a different shape or configuration), thereby adjusting the coverage angles for the loudspeaker **200**. The customizability of the waveguide insert **240** enables the loudspeaker **200** to be customized to a given space.

(20) To shape and direct the acoustic waves, the waveguide insert **240** includes a first half-horn **242**, a second half-horn **244**, and a divider **246**. As discussed in more detail below, the divider **246** can be mounted between the two horn halves **242** and **244** and extends from a region adjacent to the compression driver **222** to a mouth region of the waveguide insert **240**. The first half-horn **242** is coupled to a first longitudinal half of the frame **232** and shapes a first half of the acoustic waves. The second half-horn **244** is coupled to a second longitudinal half of the frame **232** and shapes a second half of the acoustic waves. The divider **246** acts to divide the acoustic waves generated by the compression driver **222** into halves as the acoustic waves travel from the compression driver **222**, through the waveguide **230**, and out of the mouth region.

(21) In the illustrated embodiment, the first and second half-horns **242**, **244** are asymmetrical about a primary axis (e.g., the y-axis parallel to the divider **246** in the illustrated orientation, also referred to herein as the primary plane) and symmetrical about a secondary axis (e.g., the x-axis perpendicular to the divider **246** in the illustrated orientation). More specifically, the first half-horn **242** has a wider left-right beamwidth (also referred to as a horizontal beamwidth) than the second half-horn **244**. As a result, the waveguide insert **240** will direct the first half of the acoustic waves toward a wider area to the right of the loudspeaker **200** (with reference to the direction of travel of the acoustic wave) than to the left of the loudspeaker **200**. However, because the first and second half-horns **242**, **244** have the same up-down beamwidth (also referred to as a vertical beamwidth), the waveguide insert **240** will direct the acoustic waves in the same vertical field on either side of the divider **246**. It is noted for purposes of illustration that the waveguide **230** in the embodiment illustrated in FIG. 2 that the divider **246** is positioned in a substantially vertical orientation, although the waveguide **230** and the divider **246** in other embodiments can be in a different orientation, such as a horizontal orientation or other angular orientation relative to a

vertical/horizontal frame of reference.

(22) In various embodiments, the beamwidth of a half-horn can be categorized and/or discussed by reference to the angle of coverage that the half-horn provides. The angle is measured from the longitudinal axis (e.g., the z-axis in the illustrated orientation) to the surface of the half-horn. A larger angle is associated with a wider (or larger) beamwidth. For example, in the illustrated embodiment, the first half-horn **242** has a left-right beamwidth of about 55 degrees (“°”) while the second half-horn **244** has a left-right beamwidth of about 35°, thereby establishing the wider left-right beamwidth in the first half-horn **242**. Further, both of the first and second half-horns **242**, **244** have an up-down beamwidth of about 50°, thereby establishing their equal up-down coverage. It will be understood that while the first and second half-horns **242**, **244** have been illustrated and discussed herein in the context of particular beamwidths, the waveguide insert **230** can include half-horns having any suitable left-right and/or up-down beamwidths (e.g., left-right and/or up-down beamwidths of 0°, 30°, 35°, 45°, 50°, 55°, 60°, 65°, 70°, 90°, 120°, and/or any other suitable angle).

(23) FIGS. **3A** and **3B** are exploded views of a customizable waveguide **300** configured in accordance with embodiments of the present technology. As illustrated in FIG. **3A**, the waveguide **300** is generally similar to the waveguide **230** discussed above with reference to FIG. **2**. For example, the waveguide **300** includes a driver **305**, a frame **310** coupled to the driver **305**, and a waveguide insert **320**. The frame **310** provides a rigid structure that removably receives and supports the waveguide insert **320**, the waveguide insert **320** includes asymmetric first and second half-horns **330** and **340**, and a divider **350**, which are shaped, positioned, and configured to shape and direct the acoustic waves exiting the waveguide **300**.

(24) In the embodiment illustrated in FIG. **3A**, the frame **310** includes a throat **312**, a neck **314**, and a mouth **316**. The throat **312** (sometimes also referred to as a “throat portion” and/or a “proximal end”) is operably couplable to the driver **305**. The neck **314** (sometimes also referred to as a “transition portion”) expands radially out from the longitudinal axis of the frame **310** (e.g., the z-axis). The mouth **316** (sometimes also referred to as a “mouth portion” and/or a “distal end”) is sealably couplable to a front baffle of the loudspeaker (e.g., front baffle **204** of FIG. **2**) and forms the distal end through which the acoustic waves exit the waveguide **320**. In the illustrated embodiment, the frame **310** is generally symmetrical about vertical and horizontal axes (e.g., the y-axis and the x-axis, respectively). As discussed in more detail below, the symmetry of the frame **310** can help enable quick rotation and/or other customization of the waveguide insert **320**.

(25) As further illustrated in FIG. **3A**, the first half-horn **330** can include a first throat portion **332**, a first neck portion **334**, and a first mouth portion **336**. The first half-horn **330** can be sealably coupled to a first longitudinal half of the frame **310**. For example, the first mouth portion **336** can include through-openings **339**. To sealably couple the first half-horn **330** to the frame **310**, a user can insert fasteners **360** through the through-openings **339** on the first half-horn **330** and into receiving openings **319** on the frame **310**. Once the fasteners **360** (e.g., screws, bolts, magnets, clips, and/or any other suitable element) are inserted, the first mouth portion **336** can be sealably connected to the mouth **316** and the first throat portion **332** can be sealably coupled to the throat **312**. In the illustrated embodiment, the first half-horn **330** also includes protrusions **335**. The protrusions **335** can help aid a user in coupling the first half-horn **330** (and the waveguide insert **320** more generally) by mating with grooves (e.g., ribs **524** discussed below with reference to FIG. **5**) in the frame. The protrusions **335** can also help increase the rigidity of the first half-horn **330** to reduce distortions introduced by vibrations in the first half-horn **330**.

(26) Similar to the first half-horn **330**, the second half-horn **340** includes a second throat portion **342**, a second neck portion **344**, and a second mouth portion **346**. Further, the second half-horn **340** can be sealably coupled to a second longitudinal half of the frame **310**. For example, second mouth portion **346** can include through-openings **349** that can receive the fasteners **360** in the same manner discussed above. Once the fasteners **360** (e.g., screws, bolts, magnets, clips, and/or any

other suitable element) are inserted, the second mouth portion **346** can be sealably connected to the mouth **316** and the second throat portion **342** can be sealably coupled to the throat **312**.

(27) The divider **350** is positionable between the first and second half-horns **330**, **340** to form an acoustic barrier along a primary axis therebetween. The divider **350** can be positioned in the interface between the first and second half-horns **330**, **340** to create a first acoustic travel path **338** (also referred to herein as an “acoustic pathway”) in the first half-horn **330** and a second acoustic travel path **348** in the second half-horn **340** and isolated from the first acoustic travel path **338**. As a result, acoustic waves generated by the driver **305** are split between the first and second acoustic travel paths **338**, **348** and shaped accordingly by the asymmetric first and second half horns **330**, **340**. In the illustrated embodiment, the divider **350** is captured and sealably coupled between the first and second half-horns **330**, **340**, thereby closing the open interface between the first and second half-horns **330**, **340**. The divider **350** can include edge portions **353** that contact the first and second half-horns **330**, **340** and through-openings **359** that can receive fasteners **362**. Before a user inserts the first half-horn **330** and/or the second half-horn **340** into the frame **310**, the user, for example, can insert the fasteners **362** through the through-openings **359** on the divider **350** and into receiving openings **337**, **347** on the first and second half-horns **330**, **340** (respectively). In various embodiments, the user can couple each of the components of the waveguide insert **320** together before sealably coupling the waveguide insert **320** to the frame **310** and/or can couple the first and second half-horns **330**, **340** to the frame **310** independently.

(28) In some embodiments, the customizable waveguide **300** can use first and second half-horns **330**, **340** with the same shape and contours, so the half-horns are mirror images of each other and symmetric about the vertical and horizontal axes. As a result, the beamwidths of the first and second half-horns **330**, **340** are identical. In some such embodiments, the divider **350** can be omitted since the acoustic waves do not need to be divided. In other embodiments, the shape and contour of the first half-horn **330** is different than the shape and contour of the second half-horn, so that the first and second half-horns **330**, **340** installed within the frame **310** are not mirror images of each other and can be asymmetric about one axis (e.g., asymmetric about the primary axis in the illustrated orientation) and symmetric about a second orthogonal axis (e.g., symmetric about the secondary axis in the illustrated orientation). The asymmetry along the primary axis allows the waveguide insert **320** to be customized to a space in which the loudspeaker (e.g., the loudspeaker **200** of FIG. 2) will be utilized. The symmetry along the secondary axis can help ensure that the first and second half-horns **330**, **340** form a sealed connection with the divider **350** when the waveguide insert **320** is constructed (e.g., by allowing the edges of the divider **350** to be uniformly matched to both the first and second half-horns **330**, **340**). In some embodiments, however, the first and second half-horns **330**, **340** are asymmetric about both of the primary and secondary axes. The additional asymmetry allows the waveguide insert **320** to be further customized to a space in which the loudspeaker (e.g., the loudspeaker **200** of FIG. 2) will be utilized.

(29) The modular construction of the waveguide insert **320** also allows the components and/or orientation of the waveguide **300** to be further customized. For example, as illustrated in FIG. 3B, the waveguide insert **320** can be rotated along a rotational path P.sub.1 (e.g., rotated about a longitudinal axis of the waveguide **300**). In the illustrated embodiment, the waveguide insert **320** has been rotated 90 degrees such that the vertical axis is the primary axis and the horizontal axis is the secondary axis.

(30) As discussed above, the rotation can allow a loudspeaker with a waveguide insert **320** asymmetrical about the primary axis to be customized for varying spaces. Purely by way of example, the orientation illustrated in FIG. 3B can be suitable for the loudspeakers **112** illustrated in FIG. 1 to provide a narrow beamwidth toward the periphery of the target area **102** and a wide beamwidth toward the center of the target area **102**. In another example, the orientation illustrated in FIG. 3A can be suitable for a loudspeaker carried by the ceiling of a space (e.g., the ceiling-mounted loudspeakers in a movie theater) to provide a narrow beamwidth toward the ceiling of the

space and a wide beamwidth into the rest of the space.

(31) As further illustrated in FIG. 3B, the mouth **316** of the frame **310** can have a perimeter shape symmetric about the vertical and/or horizontal axes to aid in the rotation of the waveguide insert **320**. In the illustrated embodiment, for example, the mouth **316** has a perimeter with a generally square shape (e.g., a square with rounded corners and/or convex sides) while the first and second mouth portions **336**, **346** each have perimeters matching half of the generally square shape. As a result, the waveguide insert **320** can easily be decoupled from the frame **310**, rotated by any multiple of 90 degrees about the longitudinal axis, then recoupled to the frame **310**. Further, the match in shapes between the perimeter of the mouth **316** and the first and second mouth portions **336**, **346** can help improve the seal between the frame **310** and the waveguide insert **320** by simplifying the alignment.

(32) It will be understood that, in various embodiments, the waveguide insert **320** can be broken down into additional components to further customize the waveguide insert **320**. Purely by way of example, the first and/or second half horns **330**, **340** of the waveguide insert **320** can be swapped for other half-horns with other beamwidths. In another example, the waveguide insert **320** can include quarter-horns in place of the first and second half-horns **330**, **340** illustrated in FIGS. 3A and 3B. In such embodiments, the waveguide insert **320** can include a divider that splits acoustic waves into four components, each of which is independently shaped by one of the quarter-horns (e.g., with four varying beamwidths), thereby allowing the waveguide insert **320** to be further customized to a space. Further, it will be understood that the additional customization does not require a different frame to be used in the loudspeaker. As a result, the frame **310** illustrated in FIGS. 3A and 3B can link the loudspeaker to a wide array of customized waveguide inserts.

(33) FIG. 4 is a partial cross-sectional view of a customizable waveguide **400** configured in accordance with embodiments of the present technology. In the illustrated embodiment, the customizable waveguide **400** (“waveguide **400**”) is generally similar to the waveguide **300** discussed above with reference to FIGS. 3A and 3B. For example, the waveguide **400** includes a frame **410** and a waveguide insert **420**. The frame **410** includes a throat **412**, a neck **414**, and mouth **416**. The waveguide insert **420** includes a first half-horn **430**, a second half-horn **440**, and a divider **450**.

(34) As further illustrated in FIG. 4, the throat **412** includes a driver-interface **411** operably couplable to a compression driver **405** (or other suitable acoustic driver). In various embodiments, the driver-interface **411** can be operably coupled to the compression driver **405** via one or more fasteners, adhesives, gaskets (e.g., rubber gaskets), o-rings, and/or the like. As a result of the operable coupling between the compression driver **405** and the throat **412**, acoustic waves generated by the compression driver **405** are directed into the waveguide **400**. Further, the waveguide insert **420** can be sealably coupled to the throat **412** at a first interface **422**. The seal at the first interface **422** can be created by one or more fasteners, gaskets, o-rings, and/or the like when the waveguide insert **420** is coupled to the frame **410** (e.g., using the fasteners **360** discussed above with reference to FIG. 3A). As a result of the sealed coupling between the throat **412** and the waveguide insert **420**, acoustic waves generated by the compression driver **405** are directed into the waveguide insert **420**. Still further, the waveguide insert **420** can be sealably coupled to the mouth **416** at a second interface **424**. The seal at the second interface **424** can, similarly, be created via one or more fasteners, gaskets, o-rings, and/or the like. The sealed coupling helps ensure that the acoustic waves shaped by the waveguide insert **420** are projected outward with few (or no) distortions from movement around the second interface **424**.

(35) FIG. 4 also illustrates an asymmetrical embodiment of the waveguide insert **420** that provides varying beamwidths to the resulting acoustic waves. For example, in the illustrated embodiment, the first half-horn **430** has a first pinch point **435** (sometimes also referred to herein as an “inflection point” and/or a “pivot point”) a first distance D.sub.1 from the driver-interface **411**. The first half-horn **430** is generally straight to the first pinch point **435**, then slopes peripherally

outward at first angle E (from the vertical axis) toward the second interface **424**. The first angle E defines the beamwidth of the first half-horn **430**. Similarly, the second half-horn **440** has a second pinch point **445** a second distance D.sub.2 from the driver-interface **411**. The second half-horn **440** is generally straight to the second pinch point **445**, then slopes peripherally outward at a second angle F toward the second interface **424**. The second angle F defines the beamwidth of the second half-horn **440**.

(36) As illustrated in FIG. 4, the first distance D.sub.1 is greater than the second distance D.sub.2, so the first pinch point **435** is positioned distal to the second pinch point **445** with respect to the driver-interface **411**. As a result, the first angle E is greater than the second angle F (e.g., because the first half-horn **430** must transition more quickly than the second half-horn **440**). Therefore, in the illustrated embodiment, the first half-horn **430** has a wider beamwidth than the second half-horn **440**. As discussed above, the varying beamwidths of different half-horns can be used to customize the waveguide insert **420** as suitable for a space. Purely by way of example, the narrower beamwidth of the second half-horn **440** may be used to direct acoustic waves toward the periphery of the target area **102** in FIG. 1, while the wider beamwidth of the first half-horn may be used to direct acoustic waves toward the center of the target area **102**.

(37) As further illustrated in FIG. 4, the divider **450** includes a first edge **452** (e.g., a proximal edge) and a second edge **454** (e.g., a distal edge) opposite the first edge **452**. The first edge **452** is positioned a third distance D.sub.3 from the driver-interface **411** that is non-zero. As a result, the first edge **452** is positioned a non-zero distance distally from the first interface **422**. The non-zero separation between the first edge **452** and the driver-interface **411** allows acoustic waves from the compression driver **405** to initially enter the waveguide insert **420** before being split into halves (or any other suitable split, such as thirds, quarters, eighths, and the like). The non-immediate split of the acoustic waves can improve the quality of the resulting waves. For example, by spacing the first edge **452** apart from the source of the acoustic waves, the waveguide insert **420** can reduce (or eliminate) reflections, resonance, and/or interference from the first edge **452** on the acoustic waves and/or the compression driver **405**. Larger distances between the first edge **452** and the source of the acoustic waves result in more consistent acoustic coverage for the resulting waves. However, larger distances between the first edge **452** and the source of the acoustic waves also result in larger effects from acoustic cancellation in embodiments where the waveguide insert **420** is asymmetric about one or more axes. Furthermore, the acoustic cancellation can be centered around important audio ranges (e.g., between about 1 kHz and 1.5 kHz, or around 1.1 kHz). In contrast, shorter distances between the first edge **452** and the source of the acoustic waves result in some variance in the acoustic coverage but can reduce the effects of acoustic cancellation below a range audible to the human ear. As a result, the third distance D.sub.3 can be selected to balance the tradeoffs between larger and smaller distances. In various embodiments, the third distance D.sub.3 can be between about 10 millimeters (mm) and about 50 mm, or about 16 mm.

(38) A central portion of the second edge **454** of the divider **450** is positioned a fourth distance D.sub.4 from the driver-interface **411** and a fifth distance D.sub.5 from the second interface **424**. The fourth distance D.sub.4 is greater than the first distance D.sub.1 (and the second distance D.sub.2) while the fifth distance D.sub.5 is non-zero. The second edge **454** of the divider **450** is positioned distal to the first and second pinch points **435**, **445** but within the waveguide insert **420**. The distal positioning of the second edge **454** with respect to the first and second pinch points **435**, **445** allows the divider **450** to keep the acoustic waves in the first and second half-horns **430**, **440** separate until after the acoustic waves have been shaped in their respective halves. The divider **450** can be configured to close the open interface between the first and second half-horns **430**, **440** until after the respective acoustic waves are shaped in the first and second half-horns **430**, **440**. After the acoustic waves are shaped, the interaction between the waves does not result in interference between the waves and/or affect the beamwidths of the first and second half-horns **430**, **440**. The proximal positioning of the second edge **454** with respect to the second interface **424** (and therefore

the mouth **416**) allows the acoustic waves to begin interacting before the acoustic waves interact with ambient air external of the waveguide insert **420**. The interaction can allow the acoustic waves to combine before being incident on external ambient air, thereby reducing (or minimizing) distortion on the resulting acoustic waves.

(39) FIG. 5A is a partially schematic view of a frame **500** for a customizable waveguide (e.g., the waveguide **400** of FIG. 4) configured in accordance with embodiments of the present technology. In the illustrated embodiment, the frame **500** includes a throat **510**, a neck **520**, and a mouth **530**. As discussed above, the throat **510** (sometimes also referred to as a “first end region”) is operably couplable to a compression driver **505** (or other suitable driver) to direct acoustic waves into the waveguide. FIG. 5B is a rear view of the frame **500** of FIG. 5A illustrating additional details on the throat **510**. As illustrated in FIG. 5B, the throat **510** can include radial slots **511** that help integrate the frame **500** with the compression driver **505** (FIG. 5A).

(40) Returning to FIG. 5A, the neck **520** transitions (e.g., radially expands and/or the like) from a first width at the throat **510** to a second width at the mouth **530**. In the illustrated embodiment, the neck **520** includes openings **522** that can reduce the weight of the frame **500** and/or allow air to escape as a waveguide insert is coupled to the frame **500**. In some embodiments, however, the neck **520** does not include the openings **522**. In some such embodiments, the frame **500** is sealably coupled to the compression driver **505** and/or the front baffle (e.g., front baffle **204** of FIG. 2) and/or another suitable surface. Additionally, or alternatively, the neck **520** can include ribs **524** (sometimes also referred to herein as “grooves,” “mounting tracks,” and/or the like). The ribs **524** can provide additional rigidity to the frame **500** to reduce (or minimize) distortions in the acoustic waves shaped by the waveguide. Additionally, or alternatively, the ribs **524** can help guide a waveguide insert being coupled to the frame **500**. For example, the waveguide insert can include one or more protrusions (e.g., the protrusions **335** of FIG. 3A) that nest with the ribs **524** to help guide the coupling of the waveguide insert into the frame **500** and/or help stabilize the waveguide insert once coupled to the frame **500**.

(41) The mouth **530** (sometimes also referred to as a “second end region”) is sealably couplable to an external surface of a loudspeaker (e.g., the front baffle **204** of FIG. 2) and a waveguide insert (e.g., the waveguide insert **320** of FIG. 3A). In the illustrated embodiment, the mouth **530** includes an external surface **532** that includes one or more first openings **534** (eight shown in the illustrated embodiment). The first openings **534** can receive a fastener (e.g., screws, bolts, magnets, clips, and/or any other suitable elements) to couple the mouth **530** to the external surface of a loudspeaker. The mouth **530** also includes a gasket interface **536** with one or more second openings **538**. The gasket interface **536** can help ensure that the waveguide insert is sealably coupled to the frame **500** and/or help reduce (or minimize and/or eliminate) distortions from movement at the gasket interface **536**. The second openings **538** can receive fasteners (e.g., screws, bolts, magnets, clips, and/or any other suitable elements) to couple and secure the waveguide insert to the frame **500**.

(42) As discussed above, the frame **500** can be symmetric about vertical and/or horizontal axes (e.g., the y-axis and the x-axis, respectively). The symmetry can allow the waveguide insert to be rotated between various orientations and/or allow the half-horns in the waveguide insert to be easily swapped to customize the beamwidth coverage from the resulting waveguide. In the illustrated embodiment, the external surface **532** of the mouth **530** has a generally square perimeter (in the illustrated embodiment, a square with rounded corners and slightly convex sides). In various other embodiments, the external surface **532** of the mouth **530** can have various other shapes. For example, the perimeter can be a perfect square, a generally hexagonal shape (or perfect hexagon), a generally octagonal shape (or perfect octagon), a circle, a rectangle, and/or any other suitable shape.

(43) FIG. 6 is a partially cross-sectional view of a customizable waveguide **600** configured in accordance with further embodiments of the present technology. In the illustrated embodiment, the

customizable waveguide **600** (“waveguide **600**”) is generally similar to the waveguide **300** discussed above with reference to FIGS. **3A** and **3B**. For example, the waveguide **600** includes a frame **610** and a waveguide insert **620** that includes a divider **650** captured between a first half-horn **630** and a second half-horn (not shown because of the cross-sectional view orientation).

(44) Similar to the divider **450** discussed above with reference to FIG. **4**, the divider **650** has a first edge **652** (e.g., a proximal edge) spaced apart from an interface **611** with the compression driver **605** by the third distance $D_{sub.3}$. In the illustrated embodiment, however, the first edge **652** has a concave curve (e.g., such that a central portion of the first edge **652** is spaced farther from the interface **611** than peripheral portions of the first edge **652**). The concave shape helps spread any reflections, resonance, and/or distortions caused by the split of acoustic waves at the first edge **652** across a variety of frequencies (e.g., rather than spiking around a single frequency). As a result, the reflections, resonance, and/or distortions can be imperceptible to a human ear.

(45) As further illustrated in FIG. **6**, the divider **650** has a second edge **654** (e.g., a distal edge) spaced apart from a second interface **617** between the frame **610** and the waveguide insert **620** (e.g., at the mouth of the waveguide **600**). As discussed above, the spacing between the second edge **654** and the second interface **617** helps reduce compounding negative effects (e.g., reflections, resonance, and/or distortions) by mixing the acoustic waves before they are incident on external ambient air outside of the waveguide.

(46) Further, similar to the first edge **652**, the second edge **654** has a concave curve (e.g., such that a central portion of the second edge **654** is spaced farther from the second interface **617** than peripheral portions of the second edge **654**). The concave shape helps spread any reflections, resonance, and/or distortions caused by diffraction of the acoustic waves at the second edge **654** and/or interactions as the acoustic waves rejoin to be spread across a variety of frequencies. As a result, the reflections, resonance, and/or distortions can be imperceptible to a human ear. In some embodiments, the shape of the concave curve is configured to spread the reflections, resonance, and/or distortions across the entire spectrum of frequencies generated by the compression driver **605**. The complete spread can minimize the effects at any given frequency. In some embodiments, the shape of the concave curve is configured to spread the reflections, resonance, and/or distortions across only a subset of the spectrum of frequencies generated by the compression driver **605**. The subset can be selected based on a preferred subset for the effects (e.g., rarely used frequencies, frequencies that are hard to perceive, and the like).

(47) FIG. **7** is a schematic top view of a space **701** employing a loudspeaker system **710** configured in accordance with embodiments of the present technology. In the illustrated embodiment, the space **701** includes a target area **702** (e.g., an audience zone and/or listening zone), and the loudspeaker system **710** includes three loudspeakers **712** configured to direct acoustic waves **730** toward the target zone. Further, each of the loudspeakers **712** includes a customizable waveguide **716** (referred to individually as first-third waveguides **716a-716c**) to tailor the acoustic coverage of the loudspeaker system **710** to the target area **702**.

(48) For example, the first and third waveguides **716a**, **716c** each have asymmetric beamwidths with a peripheral component at first angle G (above a front baffle of the loudspeakers **712**) and a central component at second angle H . The first angle G is greater than the second angle H . That is, the first and third waveguides **716a**, **716c** have a relatively narrow beamwidth directed at the periphery of the target area **702** and a relatively wide beamwidth directed at the central portion of the target area **702**. The relatively narrow beamwidth helps to reduce reflections from the side of the space **701** and/or cover the entirety of the periphery of the target area **702**. The relatively wide beamwidth helps cover a large portion of the central portion of the target area **702**. The first angle G (and the corresponding relatively narrow beamwidth) can be created by a first half-horn in a customizable waveguide insert while the second angle H (and the corresponding relatively narrow beamwidth) can be created by a second half-horn in the customizable waveguide insert. Further, the waveguide insert can be rotated 180 degrees between the first waveguide **716a** and the third

waveguide **716c** to create the inverted coverage pattern.

(49) Further, the second waveguide **716b** has symmetric beamwidths directed toward the central portion of the target area **702** at a third angle I. The third angle I (and the corresponding beamwidths) can be created by a third half-horn installed in the second waveguide **716b** (e.g., in place of the first and second half-horns). The second waveguide **716b** can help cover dead zones in the central portion of the target area **702** (e.g., zones that are not within the beamwidths from the first and third waveguides **716a**, **716c**) to help improve the coverage from the loudspeaker system **710**. In some embodiments, the third angle I is relatively large, thereby resulting in a narrow beamwidth from the second waveguide **716b** to cover only a small section of the central portion of the target area **702** with minimal overlap with the beamwidths from the first and third waveguides **716a**, **716c**. In some embodiments, the third angle I is relatively small, thereby resulting in a wide beamwidth from the second waveguide **716b** to cover a larger section of the central portion of the target area **702** to reduce (or eliminate) dead zones.

(50) Because the waveguides **716** are customizable ad-hoc, the loudspeaker system **710** can be tailored to any number of spaces and/or to any number of acoustic needs without needing a wide variety of the loudspeakers **712**. For example, some events (e.g., speaking events and/or conferences) may prefer to prioritize coverage of the target area **702** to reduce (or eliminate) dead zones without much emphasis on reducing overlaps. In such embodiments, the second waveguide **716b** can be given a wide beamwidth to ensure maximum coverage. Other events (e.g., movies, concerts, and the like) may prefer to minimize the number of overlaps while keeping the number of dead zones relatively low. In such embodiments, the second waveguide **716b** can be given a narrow beamwidth to cover dead zones in the target area **702** while minimizing overlap with the beamwidths from the first and third waveguides **716a**, **716c**. Because only the second waveguide **716b** needs to be altered, the loudspeaker system **710** can be quickly tailored to the varying events. Further, because the second waveguide **716b** can be removed from the loudspeaker **712**, customized, and reinserted into the loudspeaker **712**, or even replaced with yet another waveguide (not shown), to customize the coverage, the loudspeaker system **710** does not require numerous different loudspeakers **712** to be swapped in and out. As a result, the loudspeaker system **710** can be flexible between varying needs in different spaces and/or at different events.

EXAMPLES

(51) The present technology is illustrated, for example, according to various aspects described below. Various examples of aspects of the present technology are described as numbered examples (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the present technology. It is noted that any of the dependent examples can be combined in any suitable manner, and placed into a respective independent example. The other examples can be presented in a similar manner.

(52) 1. A waveguide for use in a loudspeaker having a driver and a front baffle, the waveguide comprising: a frame having a first end region positionable adjacent to the driver in the loudspeaker and a second end region opposite the first end region and positionable adjacent to the front baffle of the loudspeaker when the first end region is adjacent to the driver, wherein the second end region of the frame is symmetric about orthogonal primary and secondary axes of the frame; and a waveguide insert having: a first half-horn couplable to a first longitudinal half of the frame, the first half-horn extending from the first end region to the second end region; a second half-horn removably coupled to the first half-horn and couplable to a second longitudinal half of the frame opposite the first longitudinal half, the second half-horn extending from the first end region to the second end region; and a divider positioned between the first half-horn and the second half-horn, the divider extending from the first end region to the second end region; wherein the waveguide insert is asymmetric relative to the primary or secondary axis.

(53) 2. The waveguide of Example 1 wherein the waveguide insert is removably couplable to the frame.

- (54) 3. The waveguide of Example 2 wherein the waveguide insert is configured to be decoupled from the frame in a first orientation, rotated about a longitudinal axis of the frame while decoupled, and recoupled to the frame in a second orientation different from the first orientation.
- (55) 4. The waveguide of Example 1 wherein the first half-horn has a first acoustic beamwidth, and wherein the second half-horn has a second acoustic beamwidth different from the first acoustic beamwidth.
- (56) 5. The waveguide of Example 1 wherein the first end region of the frame is configured to be sealed with the driver when positioned in the loudspeaker.
- (57) 6. The waveguide of Example 1 wherein the divider has an edge positionable adjacent to the first end region of the frame, and wherein the edge has a concave curve spaced apart from a throat of the frame in the first end region.
- (58) 7. The waveguide of Example 1 wherein each of the first and second half-horns have a pinch point, wherein the divider has an edge positioned distal to the pinch point of each of the first and second half-horns with respect to the first end region of the frame when the divider is positioned between the first and second half-horns, and wherein the edge has a concave curve.
- (59) 8. The waveguide of Example 1 wherein the first half-horn has a first shape and the second half-horn has a second shape different than the first shape and not a mirror image of the first shape.
- (60) 9. A loudspeaker, comprising: a housing having a front baffle; a driver positioned in the housing; a waveguide frame having a first end region coupled to the driver and a second end region carried by the front baffle; and a customizable waveguide insert removably couplable to the waveguide frame and extending from the first end region to the second end region, the customizable waveguide insert comprising: a first half-horn; a second half-horn removably coupled to the first half-horn; and a divider configured to be positioned between the first half-horn and the second half-horn before the first and second half-horns are coupled together.
- (61) 10. The loudspeaker of Example 9 wherein the first half-horn defines a first acoustic pathway having a first acoustic beamwidth, and the second half-horn defines a second acoustic pathway having a second acoustic beamwidth, and wherein the divider blocked interaction between acoustic waves in the first and second acoustic pathways.
- (62) 11. The loudspeaker of Example 9 wherein the first half-horn has a first shape and the second half-horn has a second shape different than the first shape and not a mirror image of the first shape.
- (63) 12. The loudspeaker of Example 9 wherein the first half-horn has a first acoustic beamwidth, and wherein the second half-horn has a second acoustic beamwidth.
- (64) 13. The loudspeaker of Example 12 wherein the customizable waveguide insert further comprises a third half-horn couplable to the first half-horn or the second half-horn and having a third beamwidth, and wherein the customizable waveguide insert is configured to be coupled to the waveguide frame with any two of the first, second, and third half-horns.
- (65) 14. The loudspeaker of Example 9 wherein: the waveguide frame includes a throat coupled to the driver in the first end region, a mouth coupled to the front baffle at the second end region, and a neck extending distally from the throat to the mouth; and the divider has an edge positioned distal to the throat of the frame when the customizable waveguide insert is coupled to the waveguide frame, wherein the edge has a concave curve.
- (66) 15. The loudspeaker of Example 14 wherein: the first and second half-horns each have an inflection point positioned in the neck when the customizable waveguide insert is coupled to the waveguide frame; the edge of the divider is a first edge and the concave curve is a first concave curve; and the divider has a second edge positioned distal to the inflection point of each of the first and second half-horns when the customizable waveguide insert is coupled to the waveguide frame, wherein the second edge has a second concave curve.
- (67) 16. The loudspeaker of Example 14 wherein: the first half-horn has a first inflection point positioned in the neck when the customizable waveguide insert is coupled to the frame; and the second half-horn has a second inflection point positioned in the neck distal to the first inflection

point when the customizable waveguide insert is coupled to the waveguide frame.

(68) 17. The loudspeaker of Example 9 wherein the first half-horn is removable from the second half-horn, the divider, and from the waveguide frame, wherein the first half-horn is interchangeable with a third half-horn that has a shape different than the first half-horn and that is removably connectable to the second half-horn, the divider, and the waveguide frame.

(69) 18. The loudspeaker of Example 9 wherein the second half-horn is coupled to the first half-horn via one or more first removable fasteners, and the customizable waveguide insert is coupled to the frame via one or more second removable fasteners.

(70) 19. A customizable waveguide system for use in a loudspeaker, the customizable waveguide system comprising: a first partial horn extending in a longitudinal direction from a first throat portion to a first mouth portion, wherein the first partial horn comprises a first closed side and a first open side opposite the first closed side, and wherein the first partial horn is couplable to a rigid frame in the loudspeaker extending from a driver to a front baffle; a second partial horn extending in the longitudinal direction from a second throat portion to a second mouth portion, wherein the second partial horn comprises a second closed side and a second open side opposite the second closed side, and wherein the second partial horn is couplable to the first partial horn to create an open interface between the first open side and the second open side; and a divider wall couplable between the first partial horn and the second partial horn in a position to close at least a portion of the open interface between the first open side the second open side.

(71) 20. The customizable waveguide system of Example 19, further comprising: a third partial horn extending in the longitudinal direction from a third throat portion to a third mouth portion, wherein the third partial horn comprises a third closed side and a third open side opposite the third closed side, and wherein the third partial horn is couplable to (1) the first partial horn to create a second open interface between the first open side and the third open side, and/or (2) the second partial horn to create a third open interface between the second open side and the third open side, wherein: the divider wall is further couplable between (A) the first partial horn and the third partial horn in a position to close at least a portion of the second open interface, and/or (B) the second partial horn and the third partial horn in a position to close at least a portion of the third open interface.

CONCLUSION

(72) From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the technology. To the extent any material incorporated herein by reference conflicts with the present disclosure, the present disclosure controls. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Furthermore, as used herein, the phrase “and/or” as in “A and/or B” refers to A alone, B alone, and both A and B. Additionally, the terms “comprising,” “including,” “having,” and “with” are used throughout to mean including at least the recited feature(s) such that any greater number of the same features and/or additional types of other features are not precluded. Further, the terms “approximately” and “about” are used herein to mean within at least within 10 percent of a given value or limit. Purely by way of example, an approximate ratio means within a ten percent of the given ratio.

(73) From the foregoing, it will also be appreciated that various modifications may be made without deviating from the disclosure or the technology. For example, one of ordinary skill in the art will understand that various components of the technology can be further divided into subcomponents, or that various components and functions of the technology may be combined and

integrated. In addition, certain aspects of the technology described in the context of particular embodiments may also be combined or eliminated in other embodiments.

(74) Furthermore, although advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. A waveguide for use in a loudspeaker having a driver and a front baffle, the waveguide comprising: a frame having a first end region positionable adjacent to the driver in the loudspeaker and a second end region opposite the first end region and positionable adjacent to the front baffle of the loudspeaker when the first end region is adjacent to the driver, wherein the second end region of the frame is symmetric about orthogonal primary and secondary axes of the frame; and a waveguide insert having: a first half-horn couplable to a first longitudinal half of the frame, the first half-horn extending from the first end region to the second end region; a second half-horn removably coupled to the first half-horn and couplable to a second longitudinal half of the frame opposite the first longitudinal half, the second half-horn extending from the first end region to the second end region; and a divider positioned between the first half-horn and the second half-horn, the divider extending from the first end region to the second end region; wherein the waveguide insert is asymmetric relative to the primary or secondary axis.
2. The waveguide of claim 1 wherein the waveguide insert is removably couplable to the frame.
3. The waveguide of claim 2 wherein the waveguide insert is configured to be decoupled from the frame in a first orientation, rotated about a longitudinal axis of the frame while decoupled, and recoupled to the frame in a second orientation different from the first orientation.
4. The waveguide of claim 1 wherein the first half-horn has a first acoustic beamwidth, and wherein the second half-horn has a second acoustic beamwidth different from the first acoustic beamwidth.
5. The waveguide of claim 1 wherein the first end region of the frame is configured to be sealed with the driver when positioned in the loudspeaker.
6. The waveguide of claim 1 wherein the divider has an edge positionable adjacent to the first end region of the frame, and wherein the edge has a concave curve spaced apart from a throat of the frame in the first end region.
7. The waveguide of claim 1 wherein each of the first and second half-horns have a pinch point, wherein the divider has an edge positioned distal to the pinch point of each of the first and second half-horns with respect to the first end region of the frame when the divider is positioned between the first and second half-horns, and wherein the edge has a concave curve.
8. The waveguide of claim 1 wherein the first half-horn has a first shape and the second half-horn has a second shape different than the first shape and not a mirror image of the first shape.
9. A loudspeaker, comprising: a housing having a front baffle; a driver positioned in the housing; a waveguide frame having a first end region coupled to the driver and a second end region carried by the front baffle; and a customizable waveguide insert removably couplable to the waveguide frame and extending from the first end region to the second end region, the customizable waveguide insert comprising: a first half-horn; a second half-horn removably coupled to the first half-horn; and a divider configured to be positioned between the first half-horn and the second half-horn before the first and second half-horns are coupled together.
10. The loudspeaker of claim 9 wherein the first half-horn defines a first acoustic pathway having a first acoustic beamwidth, and the second half-horn defines a second acoustic pathway having a second acoustic beamwidth, and wherein the divider blocks interaction between acoustic waves in

the first and second acoustic pathways.

11. The loudspeaker of claim 9 wherein the first half-horn has a first shape and the second half-horn has a second shape different than the first shape and not a mirror image of the first shape.

12. The loudspeaker of claim 9 wherein the first half-horn has a first acoustic beamwidth, and wherein the second half-horn has a second acoustic beamwidth.

13. The loudspeaker of claim 12 wherein the customizable waveguide insert further comprises a third half-horn couplable to the first half-horn or the second half-horn and having a third beamwidth, and wherein the customizable waveguide insert is configured to be coupled to the waveguide frame with any two of the first, second, and third half-horns.

14. The loudspeaker of claim 9 wherein: the waveguide frame includes a throat coupled to the driver at the first end region, a mouth coupled to the front baffle at the second end region, and a neck extending distally from the throat to the mouth; and the divider has an edge positioned distal to the throat of the waveguide frame when the customizable waveguide insert is coupled to the waveguide frame, wherein the edge has a concave curve.

15. The loudspeaker of claim 14 wherein: the first and second half-horns each have an inflection point positioned in the neck when the customizable waveguide insert is coupled to the waveguide frame; the edge of the divider is a first edge and the concave curve is a first concave curve; and the divider has a second edge positioned distal to the inflection point of each of the first and second half-horns when the customizable waveguide insert is coupled to the waveguide frame, wherein the second edge has a second concave curve.

16. The loudspeaker of claim 14 wherein: the first half-horn has a first inflection point positioned in the neck when the customizable waveguide insert is coupled to the frame; and the second half-horn has a second inflection point positioned in the neck distal to the first inflection point when the customizable waveguide insert is coupled to the waveguide frame.

17. The loudspeaker of claim 9 wherein the first half-horn is removable from the second half-horn, the divider, and from the waveguide frame, wherein the first half-horn is interchangeable with a third half-horn that has a shape different than the first half-horn and that is removably connectable to the second half-horn, the divider, and the waveguide frame.

18. The loudspeaker of claim 9 wherein the second half-horn is coupled to the first half-horn via one or more first removable fasteners, and the customizable waveguide insert is coupled to the frame via one or more second removable fasteners.

19. A customizable waveguide system for use in a loudspeaker, the customizable waveguide system comprising: a first partial horn extending in a longitudinal direction from a first throat portion to a first mouth portion, wherein the first partial horn comprises a first closed side and a first open side opposite the first closed side, and wherein the first partial horn is couplable to a rigid frame in the loudspeaker extending from a driver to a front baffle; a second partial horn extending in the longitudinal direction from a second throat portion to a second mouth portion, wherein the second partial horn comprises a second closed side and a second open side opposite the second closed side, and wherein the second partial horn is couplable to the first partial horn to create an open interface between the first open side and the second open side; and a divider wall couplable between the first partial horn and the second partial horn in a position to close at least a portion of the open interface between the first open side the second open side.

20. The customizable waveguide system of claim 19, further comprising: a third partial horn extending in the longitudinal direction from a third throat portion to a third mouth portion, wherein the third partial horn comprises a third closed side and a third open side opposite the third closed side, and wherein the third partial horn is couplable to (1) the first partial horn to create a second open interface between the first open side and the third open side, and/or (2) the second partial horn to create a third open interface between the second open side and the third open side, wherein: the divider wall is further couplable between (A) the first partial horn and the third partial horn in a

position to close at least a portion of the second open interface, and/or (B) the second partial horn and the third partial horn in a position to close at least a portion of the third open interface.
